

Clock and reset in the TB

Most sequential DUTs need a free-running clock and a reset sequence before stimulus

Assert, hold, sync release

- Starter: active-low reset `rst_n` starts at zero while clock toggles
- Hold reset across two posedges
- The wave panel shows half-period clock phases and where reset rises relative to edges
- Compare to async release preset where `rst_n` rises between edges
- Also try hold-forever and no-reset presets to see idle and stuck-in-reset behavior

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

TB clock + reset patterns

Sketch TB **clock generation** and **reset assert / deassert** on a short timeline — how many cycles of `rst_n=0`, then release relative to `posedge clk`. Starter: hold reset two cycles, then release.

Starter example: free-running `clk`; `rst_n` low for two `posedges`, then release (classic sync-style).

Load starter example

Challenges 0 / 22 cleared

Quiz: clock: A typical TB clock is generated with...

☐ a forever toggle (or equivalent) in the testbench

☐ only the DUT's clock-gating cell

☐ \$finish inside the clock block

☐ randomize() on `clk` each ns

Show hint

Check

Next

Idle

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Core ideas

forever clk

TB clock is a free-running toggle, not a DUT output.

Assert

Drive `rst_n=0` for N clock cycles at bring-up.

Sync release

Deassert near a clock edge so flops see a clean sample.

Async release

Change reset between edges — still legal; know the hazard.

Browser lab starter

learn_sv_tb — tb clock reset

Real SV TB track practice

- In the real track, open this module's examples prompts
- Restate clock-plus-reset in one sentence
- Sketch two initial blocks on paper
- Optional: note whether your DUT expects sync or async reset from the datasheet
- No compile required yet, recognize the twin initial pattern in real testbenches

Pitfalls to watch

- Do not release reset on the wrong edge if your DUT samples sync reset on posedge clk
- Do not hold reset zero forever and wonder why nothing runs
- Do not forget clock starts defined, unknown X on clk breaks edge detection in some tools
- Mixing active-high and active-low reset names without reading the DUT causes silent
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load classic starter, count two assert posedges
- On paper, draw clk and rst_n for two cycles of assert then sync release
- When you are ready, take the short quiz, then continue to file and vector I/O